

KIROV, Russian Federation

WMO: 271990

Lat: 58.567N

Lon: 49.567E

Elev: 520

StdP: 14.42

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.5	-14.4	-25.9	1.3	-20.2	-19.2	2.0	-14.3	16.2	21.6	14.5	20.0	3.3	320	0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	84.8	68.5	81.4	66.5	78.2	64.8	70.0	81.4	68.5	78.7	66.7	75.5	4.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.3	98.6	74.6	64.7	93.2	73.1	63.1	88.0	71.9	34.3	81.8	33.0	79.3	31.5	75.5	75.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.1	12.2	11.0	-24.6	88.9	7.0	3.5	-29.6	91.5	-33.7	93.5	-37.6	95.5	-42.7	98.0		
			-24.8	72.3	6.9	2.0	-29.8	73.7	-33.8	74.9	-37.7	76.0	-42.7	77.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.0	10.4	12.7	24.5	39.3	52.8	61.0	65.6	61.0	50.6	37.8	24.3	14.8	(6)
(7)		DBStd	21.75	13.28	12.87	8.29	9.08	9.65	8.09	7.02	7.86	7.65	7.97	10.75	13.33	(7)
(8)		HDD50	5830	1226	1046	790	340	84	9	1	5	83	385	770	1091	(8)
(9)		HDD65	10052	1691	1466	1255	771	389	172	80	175	434	843	1220	1556	(9)
(10)		CDD50	1469	0	0	0	19	172	338	485	346	102	7	0	0	(10)
(11)		CDD65	215	0	0	0	0	13	50	99	51	2	0	0	0	(11)
(12)		CDH74	1779	0	0	0	3	145	407	833	377	14	0	0	0	(12)
(13)		CDH80	433	0	0	0	0	25	84	214	108	2	0	0	0	(13)
(14)	Wind	WSAvg	5.1	5.7	5.8	5.8	5.4	5.4	4.6	4.0	4.0	4.3	5.3	5.6	5.7	(14)
(15)	Precipitation	PrecAvg	24.1	1.5	0.9	1.1	1.3	1.9	2.6	3.5	2.7	2.5	1.9	1.7		(15)
(16)		PrecMax	31.5	2.8	2.2	3.0	3.2	6.2	6.5	10.0	5.0	4.8	7.2	4.5	3.1	(16)
(17)		PrecMin	16.8	0.3	0.0	0.2	0.4	0.5	0.7	0.8	0.2	0.4	0.1	0.5	0.6	(17)
(18)		PrecStd	4.2	0.6	0.4	0.7	0.7	1.2	1.5	2.1	1.1	1.1	1.3	0.9	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	35.4	45.9	71.0	83.3	86.8	88.9	89.0	75.6	63.0	47.4	39.5	(19)
(20)		MCWB	32.8	33.3	38.6	54.6	64.5	69.7	69.4	69.7	61.1	52.2	45.4	39.2		(20)
(21)		2%	DB	32.9	33.6	41.0	63.4	78.1	82.3	85.4	82.5	70.1	56.1	42.6	35.1	(21)
(22)		MCWB	31.8	32.3	35.5	50.9	60.5	66.6	68.7	68.2	59.6	50.0	41.3	33.9		(22)
(23)		5%	DB	31.2	32.2	37.6	58.0	73.6	79.0	81.9	77.7	66.1	52.0	38.9	33.1	(23)
(24)		MCWB	30.4	31.0	33.4	47.3	58.0	65.2	67.3	65.6	57.4	47.7	37.7	32.1		(24)
(25)		10%	DB	28.3	30.0	35.3	53.2	69.0	75.3	78.8	73.6	62.4	48.9	35.9	31.6	(25)
(26)		MCWB	27.3	28.6	32.3	44.8	55.3	62.7	65.6	63.6	55.7	46.3	34.8	30.9		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.2	34.0	39.3	55.6	66.6	71.5	71.9	71.6	63.7	54.5	45.5	39.2	(27)
(28)		MDCB	33.8	34.8	44.3	68.5	79.5	83.2	82.8	83.9	71.4	59.8	47.0	39.2		(28)
(29)		2%	WB	32.1	32.6	36.3	51.7	62.6	68.8	70.2	69.3	60.9	50.5	41.6	34.2	(29)
(30)		MDCB	32.8	33.4	39.7	60.8	74.7	79.6	81.2	80.8	67.8	54.4	42.6	34.8		(30)
(31)		5%	WB	30.3	31.1	34.4	48.5	59.9	66.6	69.0	67.0	58.5	48.5	37.7	32.3	(31)
(32)		MDCB	31.1	32.2	36.6	57.2	70.0	76.2	79.3	75.4	64.8	51.5	38.5	33.0		(32)
(33)		10%	WB	27.3	28.6	32.9	45.1	57.2	64.3	67.4	64.8	56.2	46.2	35.0	30.8	(33)
(34)		MDCB	28.2	29.9	35.0	52.5	66.5	72.6	76.5	72.1	61.5	48.8	35.7	31.7		(34)
(35)	Mean Daily Temperature Range	MDBR	10.0	11.7	12.9	14.8	17.8	16.7	16.7	15.1	13.1	8.5	7.1	8.7		(35)
(36)		5% DB	MCDBR	9.0	8.9	13.6	21.4	23.8	20.8	21.4	20.2	18.9	14.4	8.4	6.2	(36)
(37)		MCWBR	8.9	8.5	9.9	12.0	10.9	9.1	8.2	8.2	9.8	9.1	8.0	6.2		(37)
(38)		5% WB	MCDBR	9.0	8.9	10.9	20.1	21.2	19.2	19.1	18.6	17.0	12.9	8.0	6.5	(38)
(39)		MCWBR	8.9	8.8	8.5	11.9	10.9	9.5	8.1	8.5	9.7	9.0	7.8	6.5		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.278	0.309	0.352	0.387	0.371	0.366	0.394	0.396	0.359	0.336	0.298	0.266		(40)
(41)		taud	2.037	2.054	2.007	2.135	2.378	2.421	2.353	2.327	2.408	2.340	2.169	2.012		(41)
(42)		Ebn at Noon	162	217	242	257	273	276	264	253	246	213	157	124		(42)
(43)		Edh at Noon	17	28	39	41	34	33	35	34	27	21	15	12		(43)
(44)	All-Sky Solar Radiation	RadAvg	142	390	816	1233	1639	1743	1738	1302	767	322	115	78		(44)
(45)		RadStd	20	84	106	141	153	143	164	135	99	50	19	16		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only		+0.95	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)		+1.04	N/A	N/A	N/A	N/A	N/A	-304	N/A	N/A

Nomenclature: See separate page